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\documentclass[12pt]{article}

% Packages
\usepackage[utf8]{inputenc}
\usepackage{graphicx}
\usepackage{amsmath}
\usepackage{booktabs}
\usepackage{geometry}
\usepackage{caption}
\usepackage{hyperref}
\usepackage{natbib}

% TikZ
\usepackage{tikz}
\usetikzlibrary{shapes.geometric, arrows, positioning}

\geometry{margin=1in}

\title{Hypothesis Testing}
\author{ }
\date{April 17, 2026}

\begin{document}

\maketitle

%-----------------------------------------------
\begin{abstract}
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Hypothesis testing is a statistical technique used to evaluate assumptions about population parameters.

This paper explains hypothesis types, testing procedure, and demonstrates a real-life example.

\end{abstract}

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\section{Introduction}

Hypothesis testing is widely used in statistics to make decisions based on sample data.

It helps determine whether the observed results are due to chance or represent a real effect.

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\section{Types of Hypothesis}

\subsection{Null Hypothesis (H_0)}

Represents no change or no effect.

Example: $H_0: \mu = 170$ cm

\subsection{Alternative Hypothesis (H_1)}

Represents a significant difference.

Example: $H_1: \mu \neq 170$ cm

\subsection{Directional and Non-directional}

Directional hypotheses specify direction, while non-directional only indicate difference.

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\section{Procedure of Hypothesis Testing}

\begin{enumerate}
\item Define $H_0$ and $H_1$
\item Choose significance level ($\alpha$)
\item Select appropriate test
\item Compute test statistic
\item Calculate p-value
\item Make decision
\item Draw conclusion
\end{enumerate}

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\section{Flowchart Representation}

\begin{figure}[h]
\centering
\begin{tikzpicture}[
   node distance=1.5cm,
   every node/.style={draw, rectangle, rounded corners, align=center, minimum width=2.5cm, minimum height=1cm},
   arrow/.style={->, thick}
]

\node (start) {Hypothesis Testing};

\node (h0) [below left=of start] {$H_0$};
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\node (h1) [below right=of start] {$H_1$};

\node (alpha) [below=of h0] {Significance $\alpha$};
\node (test) [below=of h1] {Test Selection};

\node (ztest) [below left=of test] {Z-test};
\node (ttest) [below=of test] {t-test};
\node (chisq) [below right=of test] {$\chi^2$};

\node (pvalue) [below=of alpha] {Compute p-value};
\node (decision) [below=of pvalue] {Decision};

\node (reject) [below left=of decision] {Reject $H_0$};
\node (accept) [below right=of decision] {Do not Reject $H_0$};

\draw[arrow] (start) -- (h0);
\draw[arrow] (start) -- (h1);
\draw[arrow] (h0) -- (alpha);
\draw[arrow] (h1) -- (test);
\draw[arrow] (test) -- (ztest);
\draw[arrow] (test) -- (ttest);
\draw[arrow] (test) -- (chisq);
\draw[arrow] (alpha) -- (pvalue);
\draw[arrow] (pvalue) -- (decision);
\draw[arrow] (decision) -- (reject);
\draw[arrow] (decision) -- (accept);

\end{tikzpicture}
```

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\caption{Flowchart of Hypothesis Testing Process}
\end{figure}

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\section{Numerical Example}

A study claims the average height of students is 170 cm. A sample of 100 students has mean height 167 cm and standard deviation 10 cm. Test at $\alpha = 0.05$.

\textbf{Given:}

$\mu = 170,\quad \bar{x} = 167,\quad s = 10,\quad n = 100$

\subsection*{Step 1: Hypotheses}
$H_0: \mu = 170$ \\
$H_1: \mu \neq 170$

\subsection*{Step 2: Test Statistic}

Since $n > 30$, use Z-test:

\begin{equation}
Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}
\end{equation}

\subsection*{Step 3: Calculation}

\[
```

Z = \frac{167 - 170}{10/\sqrt{100}} = \frac{-3}{1} = -3

\]

\subsection*{Step 4: p-value}

\[

p \approx 0.0027

\]

\subsection*{Step 5: Decision}

Since $p < 0.05$, reject H_0.

\subsection*{Step 6: Conclusion}

There is sufficient evidence that the average height differs from 170 cm.

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\section{Summary Table}

\begin{table}[h]

\centering

\begin{tabular}{lll}

\toprule

Type & Expression & Meaning \\

\midrule

Null (H_0) & $\mu = \mu_0$ & No difference \\

Alt (Two-tailed) & $\mu \neq \mu_0$ & Any difference \\

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Alt (Right-tailed) & $\mu > \mu_0$ & Increase \\

Alt (Left-tailed) & $\mu < \mu_0$ & Decrease \\

\bottomrule

\end{tabular}

\caption{Hypothesis Types}

\end{table}


%-----------------------------------------------

\section{Conclusion}


Hypothesis testing provides a systematic framework for decision-making in statistics.

This example demonstrated how statistical tests help validate real-world assumptions.


\end{document}
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